

Supporting Information

Evaluating generative antibody–drug conjugate linkers:
chemical validity, covalent dynamics, and an FDA-anchored control

Gaël Coulombe

Hazem Mslati

This Supporting Information documents the generation pipeline (dataset construction, fine-tuning recipe, sampling sweeps) and the chemical-validity gate. The generation work is *prior* to the paper’s contribution (the evaluation result); it is reported here for reproducibility. All numbers are reported from disk-verified pipeline outputs. Section references (§) point to the main text.

Contents

S1 Dataset construction	1
S2 Fine-tuning recipe and numerical stability	2
S3 The two tradeoffs and the Pareto frontier	2
S4 Sampling sweeps and the $n=500$ re-baseline	2
S5 Chemical-validity gate specification	3
S6 Why the AMM potency oracle does not rank linkers	3

S1 Dataset construction

Payload-anchored decomposition. Each ADC in the source dataset[1] provides the full conjugate SMILES, the isolated payload SMILES, and a free-text conjugation-site annotation. Earlier pipelines guessed the linker/payload boundary with reaction SMARTS, which covered only $\approx 72\%$ of records and failed silently on non-canonical chemistries. We instead *anchor* the boundary: the provided payload is located as a subgraph of the full conjugate, and the single cross-boundary bond defines the linker/payload cut. The dataset stores the payload in its post-conjugation form, so 49/53 cysteine ADCs contain the payload as an exact substructure; an MCS fallback (RDKit `rdFMCS`, `bondCompare=CompareAny`, $\geq 70\%$ coverage) recovers the remainder. Coverage rises to **98.1 %** (52/53). When several substitution sets match, the one with the fewest junction atoms is chosen, and exactly one cross-boundary bond is required.

Deduplication and leakage control. Deduplication at the *canonical linker–payload* (L/P) level — rather than by source record UUID — caught four real train/validation leaks that a UUID-only check had masked. The cysteine-only fine-tuning set is **43 unique L/Ps** (36 in train), with **0** cross-split L/P leakage. The multi-site set extends this to **73 unique L/Ps** across three conjugation chemistries (cysteine, lysine, click).

Conformer augmentation. Each L/P is embedded in 3D with ETKDGV3 and MMFF94 optimization (UFF fallback), generating up to 20 diverse conformers per L/P (RMS pruning, minimum-diversity 0.5 Å). DiffLinker consumes a fragment file (payload + maleimide stub) and a link file (the spacer to be regenerated), with anchor indices marking the attachment atoms. The augmented cysteine-only set is 720 train / 140 validation examples; the multi-site set is larger (Table S1). Augmentation increases *geometric* diversity only; it does not add chemical (topological) diversity — 20 conformers of one molecule remain one molecule, which is the origin of the “data wall” discussed below.

Table S1. DiffLinker training sets. “L/Ps” = unique linker–payloads; “examples” = conformer-augmented fragment/link pairs after the train/validation split.

Set	unique L/Ps	examples (train/val)	chemistries
Cysteine-only (dedup)	43	720 / 140	cysteine
Multi-site	73	1240 / 202	cysteine, lysine, click

Table S2. The Pareto frontier: cysteine sampling fraction vs. connectivity and target chemistry, at the operational size (20). Connectivity and Val-Cit rates move in opposition; no configuration is dominant.

Configuration	% cysteine	native-conn. (%)	Val-Cit (%)
Multi-site (natural)	66	45	10
Curriculum (3×)	85	23	33
Cysteine-only	100	28	23

S2 Fine-tuning recipe and numerical stability

DiffLinker[2] ($E(n)$ -equivariant diffusion, pretrained on GEOM-Drugs[3]) was fine-tuned on a single RTX 4080 (16 GB). Hyperparameters: learning rate 1×10^{-5} , batch size 4, gradient clipping 1.0, *no* learning-rate warmup, *no* optimizer reset (the GEOM Adam moment estimates are restored), early stopping (patience 20) on the denoising validation loss. Best validation loss was 0.1159 for the cysteine-only fine-tune and 0.1143 for the multi-site fine-tune.

The isfinite guard. Large ADCs ($\gtrsim 95$ heavy atoms) at high diffusion timesteps occasionally produce a non-finite forward loss in single precision (a physics/numerics overflow, $\approx 2\%$ of batches). A pre-backward **isfinite** guard that skips the offending batch — combined with the restored Adam moments — prevents a NaN cascade. In a controlled ablation (5 runs $\times$ 200 steps), the guard gave 0/5 NaN cascades and 5/5 convergence; without it, 4/5 runs diverged. Resetting the optimizer or adding warmup *worsened* stability, because the restored GEOM moments actively pull weights out of the overflow region.

Denoising loss is not generation quality. The validation loss is a diffusion *denoising* objective and is *not* a direct measure of generated-molecule quality. We therefore report generation quality with the operational usable-3D yield and Wilson 95 % confidence intervals (Section S4), not with validation loss.

S3 The two tradeoffs and the Pareto frontier

Two intrinsic tradeoffs constrain the generator; neither is removed by any easy lever, so the operating point can only be *moved along* a Pareto frontier.

Chemistry vs. connectivity (the size axis). Within one model, forcing larger linkers increases ADC chemistry but decreases geometric cohesion. As the forced linker size rises from 15 to 60, single-fragment connectivity falls ($\approx 30\% \rightarrow \approx 4\%$) while the Val-Cit and urea motif rates rise (to 64 % and 81 % respectively; Table 1, main text). The operational size is chosen near the knee (20–25).

Specificity vs. diversity (the dataset axis). Broadening the training distribution (multi-site) improves connectivity but dilutes the dominant cysteine–Val-Cit signal toward the population mean. The two effects trade off monotonically with the *cysteine sampling fraction*, which therefore acts as an operational slider along the frontier (Table S2). No setting maximizes both connectivity and target chemistry.

S4 Sampling sweeps and the $n=500$ re-baseline

Usable-3D definition. A generated molecule is *usable-3D* if it is (a) natively connected (a single fragment under distance-based bond perception, with *no* post-hoc minimum-spanning-tree closure), (b) MMFF94-valid

Table S3. $n=500$ re-baseline (3D-strict), Wilson 95 % CI. The $\lambda=0.01$ arm and the $\lambda=0$ control have disjoint CIs despite the penalty being a no-op in the control — the difference is run-to-run training variance, not a real effect. Curriculum, multi-site, and the $\lambda=0$ control are mutually overlapping.

Config ($n=500$)	conn. %	Val-Cit %	usable-3D %	95 % CI
Curriculum (baseline)	20.8	6.6	8.6	[6.4, 11.4]
Multi-site (base)	40.0	4.4	4.8	[3.2, 7.0]
$\lambda=0.01$ (prev. best)	33.2	9.4	10.2	[7.8, 13.2]
$\lambda=0$ control	15.4	4.4	4.6	[3.1, 6.8]

(radical-neutralized, sanitizable, and minimizable to a finite energy while remaining a single fragment), and (c) ADC-motif-bearing (Val-Cit or urea). This is the gate of §4.2; it is the strict 3D criterion required for the downstream covalent build, which a 2D-only screen would not satisfy.

Re-baseline. Early single-checkpoint comparisons at $n=100$ suggested a $\approx 1.5\text{--}2\times$ yield gain from 50 extra fine-tuning epochs. Re-running at $n=500$ with Wilson 95 % confidence intervals (Table S3) erased it: a clean $\lambda=0$ control — in which the additional connectivity penalty is a mechanical no-op — regressed to baseline rather than matching the $\lambda=0.01$ arm, with *disjoint* confidence intervals. The only difference between the two runs is GPU-nondeterministic training-trajectory variance. **The apparent gain was a single favourable-variance draw.** The operational yield is $\approx 9\text{--}10$ usable-3D candidates per 100. The methodological consequence: single-checkpoint comparisons at $n<500$ cannot separate variance from signal; multi-seed evaluation with confidence intervals at $n\geq 500$ is required.

Novelty. Across 1000 generations there were **zero** near-copies (Tanimoto > 0.85 to any training L/P), and the median Tanimoto to the training set was 0.44–0.59 — novel molecules on a shared payload scaffold, not memorization.

S5 Chemical-validity gate specification

The gate runs fail-closed in two levels (§3.2).

Level A — ligand (RDKit[4]).

1. Sanitizable (valence-consistent Kekulé/aromaticity perception).
2. Single connected component (rejects the diffusion fragmentation that dominates raw output).
3. No undefined stereocenters on linker or payload (the coordinate $\rightarrow$ SDF conversion strips stereochemistry; canonical payload stereochemistry is reimposed by CIP-driven iterative chiral-tag assignment and verified from the 3D coordinates).
4. Correct maleimide vs. thiosuccinimide oxidation state (open maleimide $\text{O}=\text{C}_1\text{C}=\text{CC}(=\text{O})\text{N}_1$ distinguished from the conjugated thiosuccinimide).
5. Basic drug-likeness (molecular weight, logP, rotatable bonds, TPSA within ranges).

Level B — conjugate topology. A text-level parse of the assembled GROMACS topology enforces a per-atom maximum-valence table (carbon ≤ 4 , sulfur ≤ 2 for thioether, etc.). This is the guard that auto-catches the over-valent conjugation carbon of §4.4 and the hypervalent-sulfur defect (a retained cysteine thiol proton) — both of which pass a distance-only check, because a harmonic bond holds its length regardless of valence. The lesson (Table 2, main text): validate the reacting atom’s bond *count and type*, not just the new bond’s *length*.

S6 Why the AMM potency oracle does not rank linkers

The AMM classifier[1] predicts whether an ADC is cytotoxic below a 10 nM threshold from a 2D conjugate representation plus biological context. It does not rank the generated linkers, for two compounding reasons.

(i) **Label saturation:** a potent payload (MMAE) on an amplified antigen (HER2) at a 10 nM threshold is predicted active for essentially any reasonable linker. (ii) **No applicability-domain check:** after a silent checkpoint-loading defect was corrected (a vestigial untrained head had to be dropped and only the attention keys remapped; diagnosed via BatchNorm tracking counters), the genuine model still scores chemically irrelevant “junk” control molecules (aspirin, caffeine, ethanol) as active — their molecular features are out-of-distribution, propagating to extreme logits. AMM is therefore usable only as a coarse plausibility filter (real-vs-junk), never as a fine-grained ranker. The discriminating signal — linker-driven plasma stability — was never in its training data.

References

- [1] Hazem Mslati et al. ADCpedia: a multimodal model for antibody–drug conjugate potency prediction. *bioRxiv*, 2026. doi: 10.64898/2026.05.22.727320. Preprint.
- [2] Ilia Igashov, Hannes Stärk, Clément Vignac, Arne Schneuing, Victor Garcia Satorras, Pascal Frossard, Max Welling, Michael Bronstein, and Bruno Correia. Equivariant 3D-conditional diffusion model for molecular linker design. *Nature Machine Intelligence*, 6:417–427, 2024. doi: 10.1038/s42256-024-00815-9.
- [3] Simon Axelrod and Rafael Gómez-Bombarelli. GEOM, energy-annotated molecular conformations for property prediction and molecular generation. *Scientific Data*, 9:185, 2022. doi: 10.1038/s41597-022-01288-4.
- [4] RDKit. RDKit: Open-source cheminformatics. <https://www.rdkit.org>. Accessed 2026.